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Hello!

Amazon SQS

MUHAMMAD NEGM

What's Amazon Simple Queue ?

A message queue service offers reliable and scalable hosted queues for storing messages as they travel between computers.

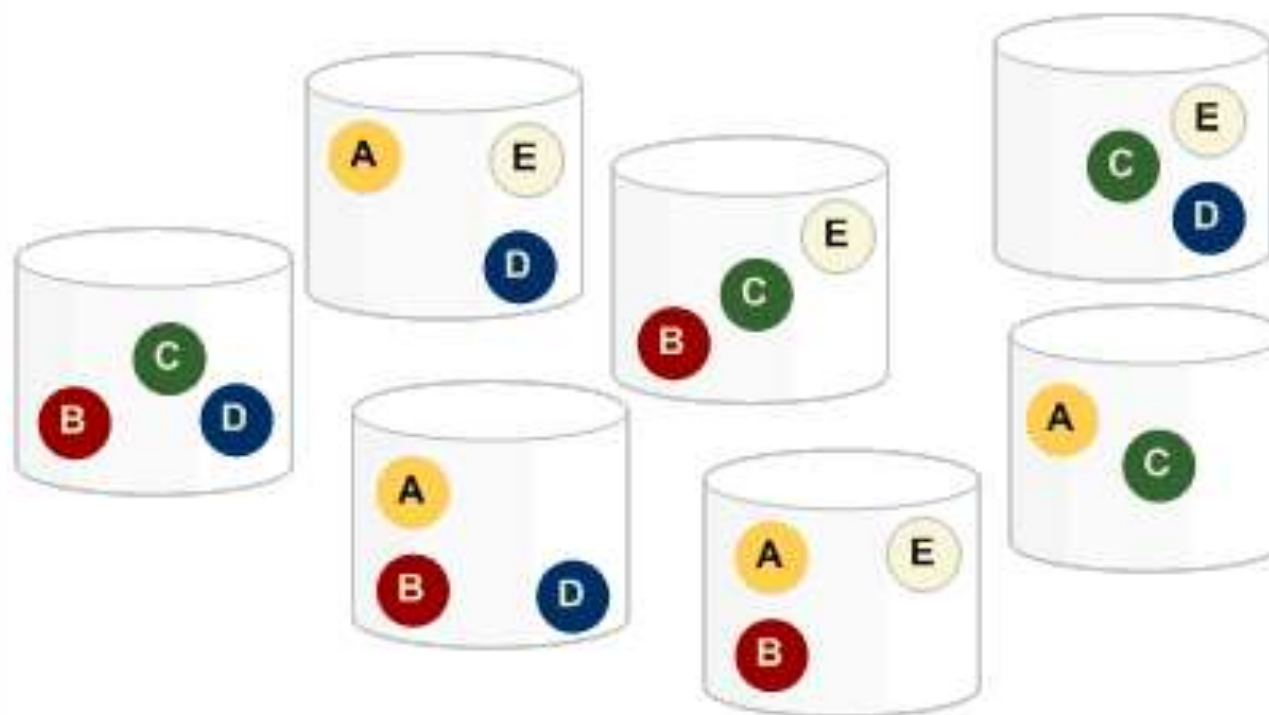
Your Distributed System's Components

Component
1

Component
2

Component
3

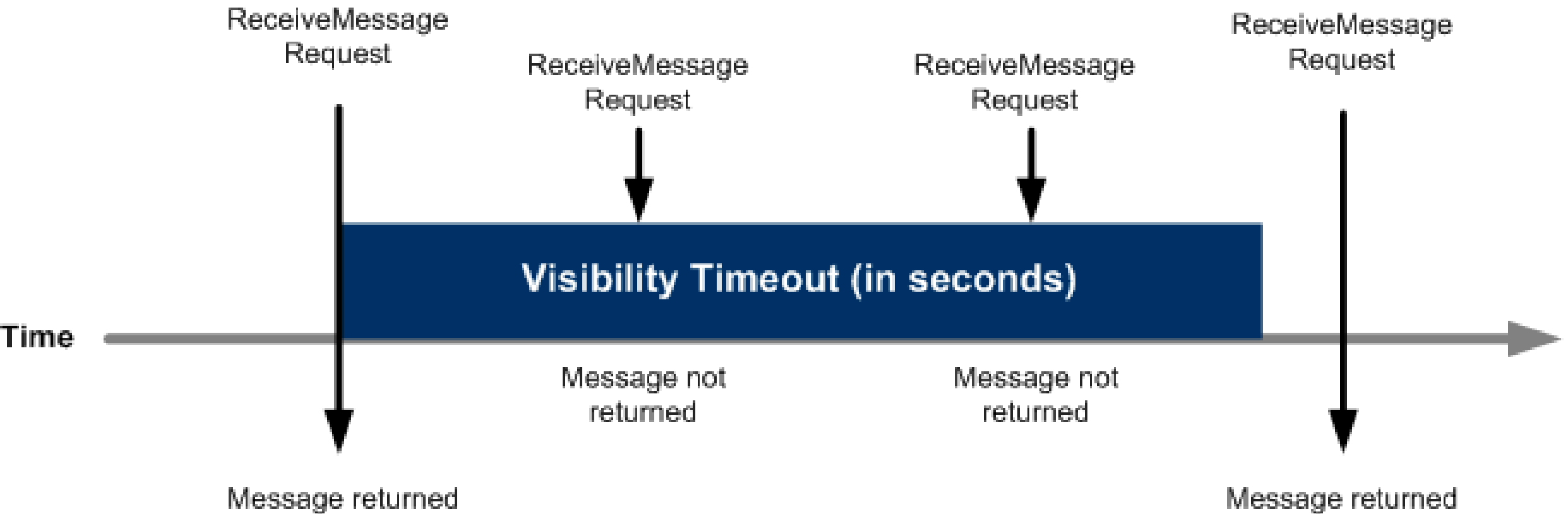
Your Queue (Distributed on SQS Servers)



Why Amazon SQS ?

By using Amazon SQS, you can move data between distributed components of your applications that perform different tasks without losing messages or requiring each component to be always available.

Life Cycle



Advantages of message queues:

Asynchronous: Queue it now, run it later.

Decoupling: Separates application logic.

Resilience: Won't take down your whole application if part of it fails.

Redundancy: Can retry jobs if they fail.

Guarantees: Makes sure that jobs will be processed.

Scalable: Many workers can process individual jobs in a queue.

Profiling: Can aid in identifying performance issues.

Disadvantages of message queues:

Asynchronous: you have to wait until a job is complete.

Load: each job in the queue must wait its turn before it can be processed. If one job overruns, it affects each subsequent job.

Architecture: the application needs to be designed with queues in mind.

Use cases of message queues:

Sending/receiving data from a third-party APIs

Sending an e-mail

Generating reports

Example Queues

IronMQ

Amazon SQS

RabbitMQ

Beanstalkd

Communication Protocols

PUSH

RESTFUL

Pubsubhubbub

Other 3rd Party protocols



That's all Folks!